SCIENCE 10

Practice Problems for Exam

1. Complete each of the following equations by adding the missing species:

$$^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \longrightarrow \, ? + {}^{1}_{0}\mathrm{n}$$

$$^{239}_{94}\mathrm{Pu} + \, ? \longrightarrow {}^{242}_{96}\mathrm{Cm} + {}^{1}_{0}\mathrm{n}$$

$$^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \longrightarrow \, ? + {}^{1}_{1}\mathrm{H}$$

$$^{235}_{92}\mathrm{U} \longrightarrow \, ? + {}^{135}_{55}\mathrm{Cs} + 4\,{}^{1}_{0}\mathrm{n}$$

2. Arrange the following elements in order of **increasing atomic size**:
   Na, Al, Si, P
3. Arrange the following in order of **increasing electronegativity**:
   Li, F, C, O
4. Complete the table below.

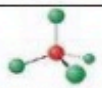
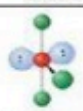
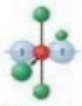
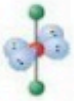
Molecule	Lewis Structure	VSEPR Notation	Molecular Geometry	Type of IMFA
SO_2				
CCl_4				
PCl_3				
XeF_2				
H_2S				
NF_3				
BrF_5				

5. Arrange each of the following sets of compounds in order of increasing viscosity:
   a. HCl, H_2O, SiH_4
   b. F_2, Cl_2, Br_2
   c. CH_4, C_2H_6, C_3H_8

B = Solids Hg = Liquids Kr = Gases Pm = Not found in nature

1																	18
1 H 1.00794	2											13	14	15	16	17	2 He 4.002602
3 Li 6.941	4 Be 9.012182											5 B 10.811	6 C 12.0107	7 N 14.00674	8 O 15.9994	9 F 18.9984032	10 Ne 20.1797
11 Na 22.989770	12 Mg 24.3050	3	4	5	6	7	8	9	10	11	12	13 Al 26.581538	14 Si 28.0855	15 P 30.973761	16 S 32.066	17 Cl 35.4527	18 Ar 39.948
19 K 39.0983	20 Ca 40.078	21 Sc 44.955910	22 Ti 47.867	23 V 50.9415	24 Cr 51.9961	25 Mn 54.938049	26 Fe 55.845	27 Co 58.933200	28 Ni 58.6534	29 Cu 63.545	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.92160	34 Se 78.96	35 Br 79.504	36 Kr 83.80
37 Rb 85.4678	38 Sr 87.62	39 Y 88.90585	40 Zr 91.224	41 Nb 92.90638	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.90550	46 Pd 106.42	47 Ag 196.56655	48 Cd 112.411	49 In 114.818	50 Sn 118.710	51 Sb 121.760	52 Te 127.60	53 I 126.90447	54 Xe 131.29
55 Cs 132.90545	56 Ba 137.327	71 Lu 174.967	72 Hf 178.49	73 Ta 180.94.79	74 W 183.84	75 Re 186.207	76 Os 190.23	77 Ir 192.217	78 Pt 195.078	79 Au 196.56655	80 Hg 200.59	81 Tl 204.3833	82 Pb 207.2	83 Bi 208.58038	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	103 Lr (262)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Cn (277)	113 Uut (277)	114 Uuq (277)	115 Uup (277)	116 Uuh (277)		118 Uuo (277)

57 La 138.9055	58 Ce 140.116	59 Pr 140.50765	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.964	64 Gd 157.25	65 Tb 158.92534	66 Dy 162.50	67 Ho 164.93032	68 Er 167.26	69 Tm 168.93421	70 Yb 173.04
89 Ac 232.0381	90 Th 232.0381	91 Pa 231.035888	92 U 238.0289	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)

Number of Electron Dense Areas	Electron-Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
2 	Linear	Linear 				
3 	Trigonal planar	Trigonal planar 	Bent 			
4 	Tetrahedral	Tetrahedral 	Trigonal pyramidal 	Bent 		
5 	Trigonal bipyramidal	Trigonal bipyramidal 	Sawhorse 	T-shaped 	Linear 	
6 	Octahedral	Octahedral 	Square pyramidal 	Square planar 	T-shaped 	Linear 

Answers:

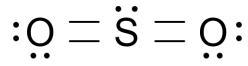
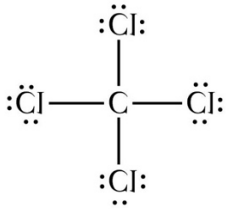
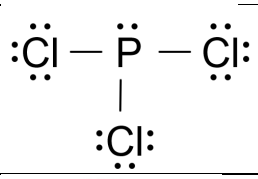
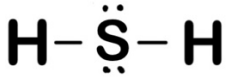
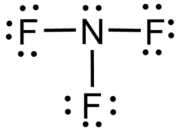
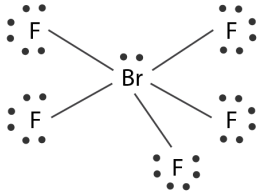
1. $^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \longrightarrow ? + {}^{1}_{0}\mathrm{n}$ $^{30}_{15}\mathrm{P}$

   $^{239}_{94}\mathrm{Pu} + ? \longrightarrow {}^{242}_{96}\mathrm{Cm} + {}^{1}_{0}\mathrm{n}$ $^{4}_{2}\mathrm{He}$

   $^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \longrightarrow ? + {}^{1}_{1}\mathrm{H}$ $^{17}_{8}\mathrm{O}$

   $^{235}_{92}\mathrm{U} \longrightarrow ? + {}^{135}_{55}\mathrm{Cs} + 4\,{}^{1}_{0}\mathrm{n}$ $^{96}_{37}\mathrm{Rb}$

2. Al < Si < P < Na
3. Li < C < O < F
4.

Molecule	Lewis Structure	VSEPR Notation	Molecular Geometry	Type of IMFA
SO_2		AX_2E_1	bent	polar
CCl_4		AX_4	tetrahedral	nonpolar
PCl_3		AX_3E_1	trigonal pyramidal	polar
H_2S		AX_2E_2	bent	polar
NF_3		AX_3E_1	trigonal pyramidal	polar
BrF_5		AX_5E_1	square pyramidal	polar

5. Arrange each of the following sets of compounds in order of increasing viscosity:

a. HCl, H_2O, SiH_4
   SiH_4 (nonpolar, only has LDF), HCl (polar; LDF and DDI), H_2O (polar; LDF, DDI, hydrogen bond)
b. F_2, Cl_2, Br_2 (higher atomic number/number of electrons, stronger IMFA)
c. CH_4, C_2H_6, C_3H_8 (more carbon, stronger IMFA)